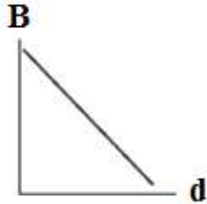
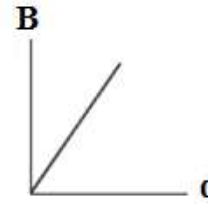
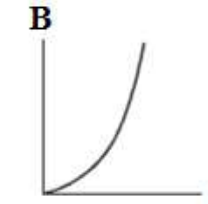
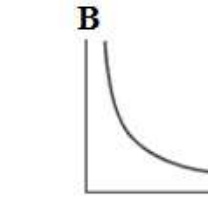


Answer the following groups**Group 1 :****Write the scientific term expressed by the following statements:**

	Statement
1	It is the magnetic flux density which exerts a force of 1 N on a wire of 1m length carrying a current of intensity 1A placed perpendicularly to the magnetic field
2	They are induced currents that circulate in closed paths due to the change in magnetic flux through a solid conductor associating with heating effect
3	The phenomenon of inducing emf in a coil due to change in current in the same coil and hence the change in magnetic flux in the coil
4	The ratio of output electric power to the input electric power
5	It is an ideal system that absorbs all radiation falling on it. Then re-emit this radiation totally again
6	It is the attractive force between the positive ions and free electrons in metal which make the electrons cannot leave it
7	It is the minimum energy needed to free (release) an electron from the surface of the metal
8	They are invisible electromagnetic waves of short wavelengths, located between gamma rays and UV rays
9	An external photon of energy $E_2 - E_1$ passes by excited atom at E_2 to relax it to E_1 before life time is over
10	The circuits that perform logic operations and depend on digital electronics

Group 2 :**Put circle around the correct answer**

11	When the density of magnetic field in a given area is increased to its double value. The magnetic flux through this area would be..... (a) unchanged (b) doubled (c) increased four times (d) increased eight times
12	Which the graph correctly gives the relation between the magnitude of the magnetic field density (B) outside an infinitely long straight current-carrying wire and the distance of point from the wire (d)?  (a)  (b)  (c)  (d)
13	A solenoid is connected to a battery in a closed circuit, if its turns are compressed regularly. The magnetic flux density at a point on its axis inside it will (a) increase (b) decrease (c) remain the same (d) vanish
14	6) A length of straight conductor, carrying a current of 3A perpendicular to a magnetic field of 0.5 T, experiences a force of 1.5 N. What is the length of the conductor? (a) 0.5 m (b) 1 m (c) 2 m (d) 4 m
15	A simple AC generator is made by rotating a flat rectangular coil in a uniform magnetic field. The instantaneous value of emf produced when the coil is parallel to the magnetic field is (100 V). If the frequency of rotation is doubled, then the instantaneous value of emf produced at the same position is..... (a) 50V (b) 100V (c) 200V (d) 400V
16	A black body is heated to a temperature of 3000 K so the wavelength that is associating with its radiation of the maximum intensity was (λ). If it is cooled to an absolute temperature (T) then the wavelength associating with its radiation of the maximum intensity became (10λ) so the temperature (T) is (a) 300 K (b) 2700 K (c) 270 K (d) 1800 K
17	Which of the following statements is correct when connecting the diode in reverse direction? (a) The width of depletion layer increases and diode resistance decreases. (b) The width of depletion layer decreases and diode resistance decreases. (c) Neither the width of depletion layer nor the diode resistance changes. (d) The width of depletion layer increases and the diode resistance increases.

18	All of the following are from the properties of photons <u>Except</u>..... (a) they travel in straight line with constant speed equals the speed of light. (b) they have linear momentum(P_L). (c) they can be deflected under the effect of electric and magnetic fields. (d) they travel in space in the form discrete packets of energy (quanta).
19	A monochromatic light falls on a metallic surface, the electrons with kinetic energy are emitted when (a) the frequency of incident photon decreases (b) the energy of the incident photon is greater than the work function of the metal (c) the energy of the incident photon is less than the work function of the metal (d) the energy of the incident photon equal to the work function of the metal
20	In an ideal step up transformer, which of the following statements is <u>incorrect</u> (a) The current ratio of secondary to primary is smaller than one. (b) The turn's ratio of primary to secondary is smaller than one. (c) The voltage ratio of primary to secondary is greater than one. (d) The output-to-input electric power ratio is smaller than one.

Group 3 :**Correct the underlined in the following statements**

	Statement
21	The maximum value of magnetic flux cutting a certain area when the magnetic flux lines is <u>parallel to</u> that area
22	The magnetic flux density at center of a circular coil carrying a current decreases by <u>decreasing</u> the cross sectional area of the coil
23	The magnetic dipole moment $ md $ is given by the following relation $ md = \underline{IAN\sin\theta}$
24	The mutual induction coefficient measured in <u>Ampere/second</u>
25	The wavelength of the scattered photon ($\lambda_{\text{scattered}}$) is <u>smaller than</u> the wavelength of the incident photon ($\lambda_{\text{incident}}$)
26	The wavelength of the characteristic radiation of X- rays <u>increases</u> as the atomic number of target material increases
27	The ordinary light is <u>not subject</u> to the inverse square law
28	<u>Decrease</u> the temperature is one of the methods to increases the electric conductivity of semiconductors.

Group 4 :**Answer the following****problem**

29 When the current in a coil changes from 5A to 2A in 0.05 s, an e m f of 15 V is
30 induced in the coil. **Calculate the coefficient of self-induction of the coil.**

31 A rectangular coil of 500 turns has dimensions 0.2m × 0.1 m, carries a current 5A. The
32 plane of the coil makes angle 60° to a uniform magnetic field of flux density 0.8 T.
Calculate the magnetic torque acting on the coil

33 A metallic ring is placed in a uniform magnetic field perpendicular to the plane of the
34 ring. If the magnetic flux changes from 0.05 Wb to – 0.05 in 0.05 s. **Calculate the
average induced EMF in the loop**

35 A metal has a work function 3.85×10^{-19} J. **Calculate the maximum kinetic energy of**
36 **electrons emitted from the metal** when light of frequency 7.5×10^{14} Hz is shone on the
surface (Given that Planck's constant (h) = 6.6×10^{-34} J.s)

37 The current gain (β) of a transistor in common emitter mode is 50. **Calculate the**
38 **collector current if the base current 50μ A**

Group 5:

Mention one factor only that affects each of the following:

	Statement
39	The density of magnetic field at a point along the axis of solenoid carrying current.
40	The magnetic torque acting on a rectangular coil carrying current placed parallel in a magnetic field
41	The mutual induction coefficient between two coils
42	The induced emf in a straight wire moving normal to a magnetic field
43	The velocity of electron emitted from the cathode in cathode ray tube
44	The minimum wavelength in the continuous radiation of X-rays

Group 6:**Compare between each of the following**

45	Point of comparison	NOR gate				NAND gate				
46		Complete the truth table	Input A	Input B	Output		Input A	Input B	Output	
			1	1			1	1		
47	Point of comparison	Direct current (DC).				Alternating current (AC)				
48		The direction								
49	Point of comparison	Lenz’s rule				The right Cork Screw rule				
50		The use								

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